FUNDAMENTALS OF HEALTH DATA

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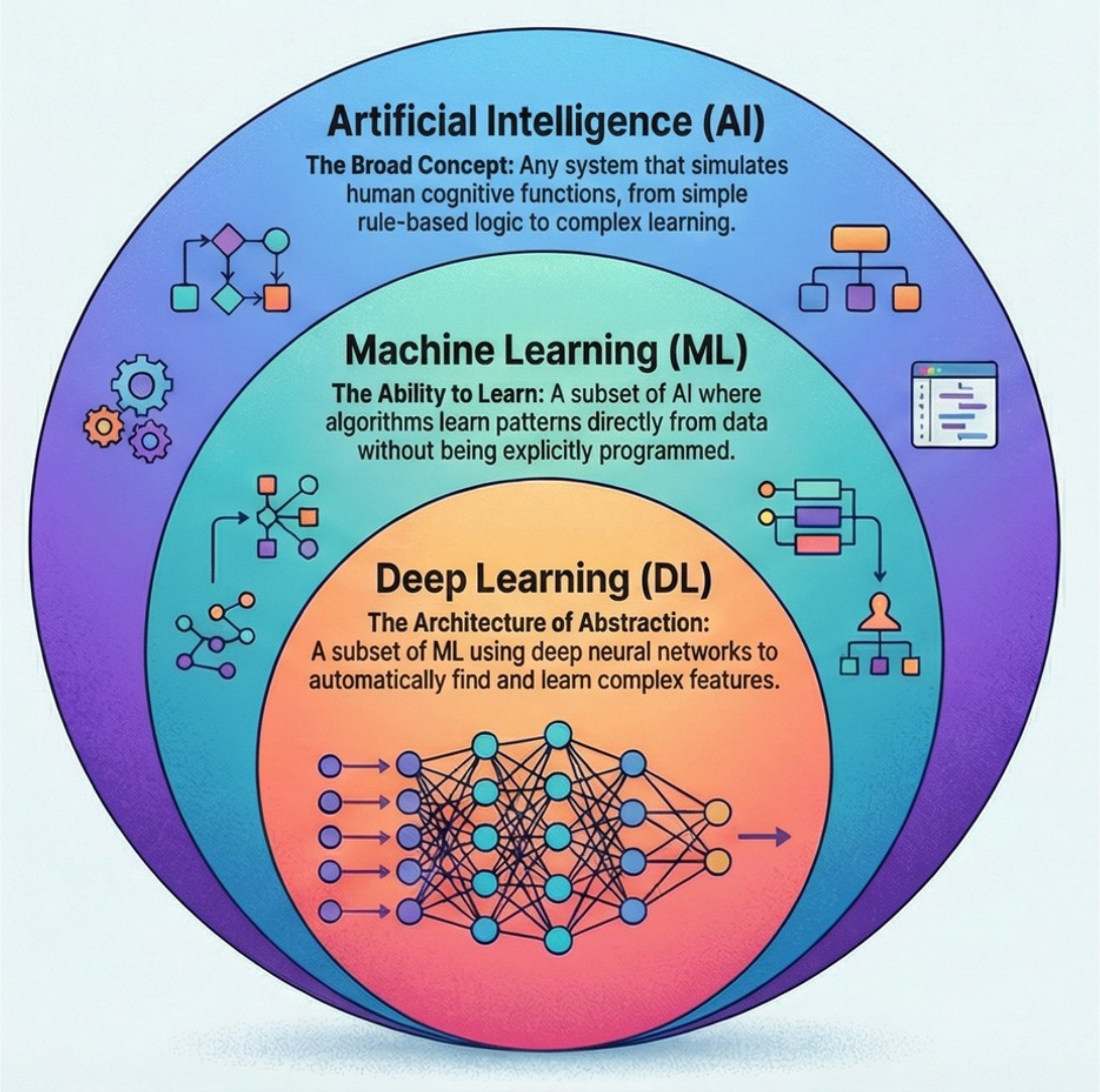
Introduction

- Artificial intelligence (AI) and machine learning (ML) have advanced various fields of research, particularly in healthcare.
- AI describes computational programs that simulate human intelligence.
- ML is a subset of AI that extracts patterns from data automatically.
- Deep learning (DL) is a subset of ML that can automatically learn features.



ML in healthcare

Predictive analytics	• Analyzes health data to predict risks such as hospital readmission or disease onset, enabling early intervention.
Diagnosis and treatment	• Interprets medical images and clinical data to support accurate diagnosis and guide appropriate treatment choices.
Personalized medicine	• Tailors treatment decisions by using individual patient characteristics like genetics and medical history.
Clinical decision support	• Integrates into clinical systems to help healthcare providers make more informed and timely decisions.
Population health management	• Identifies patterns in large datasets to inform public health strategies and improve health outcomes at the population level.

Designing a ML model

Problem definition

- Identify problems and challenges in the healthcare field.

Dataset

- ML models rely on diverse datasets collected from multiple platforms.

Data pre-processing

- Cleans and improves data quality so that models can achieve more accurate and reliable performance.

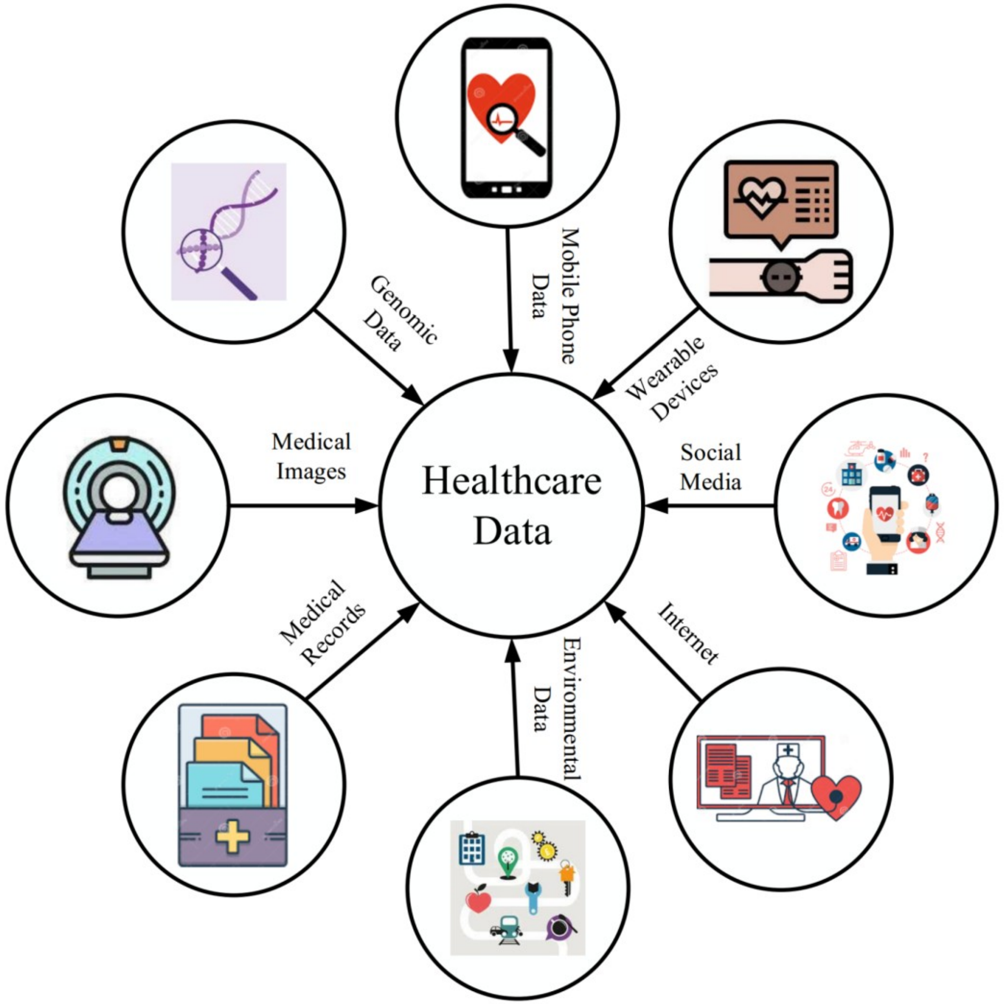
ML model development

- Balance model complexity, training time, and inference speed with dataset size, while selecting an appropriate learning approach to ensure both accuracy and practical usability

Evaluation

- Data quality, model design and performance to ensure the model behaves as expected, is reliable for real-world deployment, and continues to perform effectively after implementation.

Types of healthcare data



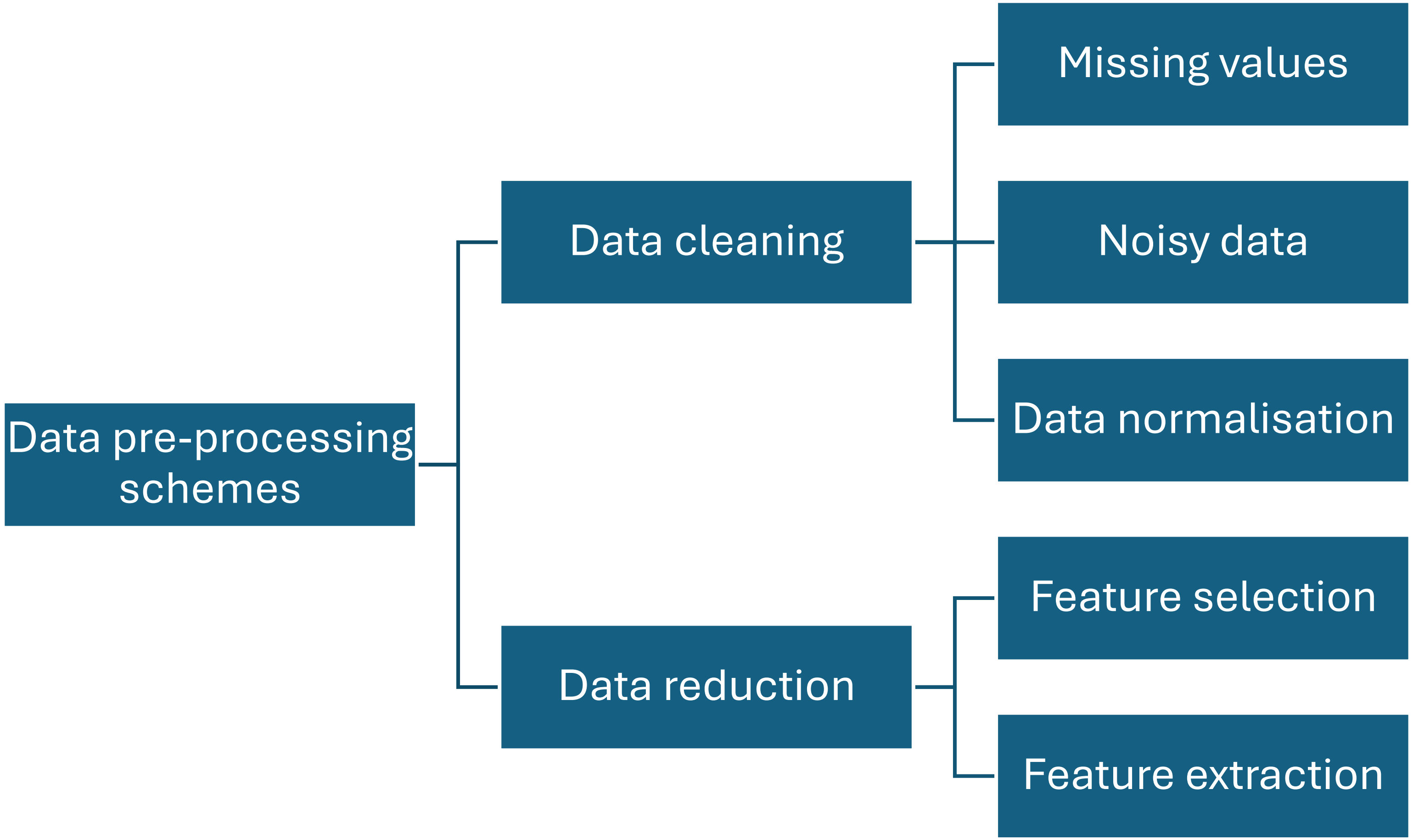
Source: Rahmani, A. M., et al. (2021). https://doi.org/10.3390/math9222970

- Healthcare data can be derived from common sources such as health surveys, clinical trials and patient registries.
- They can also include medical records and be patient-generated through wearable devices.
- Different sources means different formats.

Types of healthcare data

Data type	Characteristics	Examples
Structured	Highly organized, easily searchable, and formatted into standardized fields (rows and columns) within relational databases.	Patient demographics, numeric lab values, blood pressure readings, medication dosages, and ICD-10 billing codes
Semi-structured	Partially organized, combining both structured and unstructured.	FHIR resources, clinical emails, JSON laboratory outputs
Unstructured	Lacks a uniform format. Cannot be easily queried using traditional databases.	Radiology reports, discharge summaries, clinical consultation notes, social and behavioural descriptions
Image data	Visual information represented in a digital format, typically organized as a grid of numbers known as pixels.	X-rays, CT scans, MRIs, ultrasounds
Biosignals	Continually measured physiological signals from living organisms that communicate the body's electrical, mechanical, or chemical activity	Electrocardiogram (ECG/EKG), heart rate (HR), respiratory rate (RR), blood glucose levels

Data pre-processing



Missing data

MCAR (Missing Completely at Random)

- Missing values occur randomly
- Example: A lab test result lost due to system error

MAR (Missing at Random)

- Missingness depends on observed data
- Example: Younger patients less likely to have certain tests recorded

MNAR (Missing Not at Random)

- Missingness depends on the missing value itself
- Example: Patients with high viral load avoid testing

MICE (Multiple Imputation by Chained Equations)

- Fills in missing values multiple times
- Creates several complete datasets
- Accounts for uncertainty which means more reliable results

Indicator Variable Approach

- Create a new variable indicating whether a value is missing
- Example: Viral load missing = 1 (missing), 0 (observed)
- Useful when missingness itself is informative

Noisy data

- Filtering methods, e.g. smoothing techniques, are used to remove noise in health datasets. This improves the accuracy of the learning model.
- However, the detection of noisy data is not easy.
- A solution is to examine the database by professionals and physicians to improve its quality.
- This causes more accurate modeling and reduces its error.
- However, this work is costly and time-consuming.

Normalisation

- Ensures variables are on comparable scales
- Prevents large-scale variables from dominating models

Z-score scaling	Min-max scaling	Robust scaling (IQR-based)	Per-channel ImageNet normalisation (CNNs)
• Centers data around mean = 0, SD = 1 • Most common for statistical models	• Scales values between 0 and 1 • Often used in neural networks	• Uses median and interquartile range • More stable when outliers exist	• Standardises image inputs using precomputed mean and SD • Used when applying deep learning to medical images

Class Imbalance

- Occurs when one outcome is much rarer than another
- Example: Few patients with treatment failure vs many stable patients

SMOTE (Synthetic Minority Oversampling Technique)	Class-weighted loss	Threshold calibration	Under-sampling
• Creates synthetic examples of the minority class • Helps balance the dataset	• Assigns higher penalty to misclassifying rare cases • Common in logistic regression or neural networks	• Adjusts decision threshold (not always 0.5) • Example: classify as "high risk" at lower probability	• Reduce number of majority class observations • Risk: losing useful information

Feature engineering

- Creating new variables to improve model performance

Interaction terms

- Combine variables to capture relationships
- Example: age × CD4 count

Polynomial features

- Capture nonlinear relationships
- Example: age^2 or biomarker combinations

Dimensionality reduction (PCA / UMAP)

- PCA: reduces variables while preserving variance
- UMAP: preserves structure for clustering/visualization

Feature selection

- Feature selection is the process of choosing the most relevant variables from a dataset to improve model performance, reduce complexity, and eliminate unnecessary or noisy features.

Wrapper methods

- Evaluate different subsets of features by repeatedly training a model and selecting the subset that yields the best performance.
- Examples: forward selection, backward selection

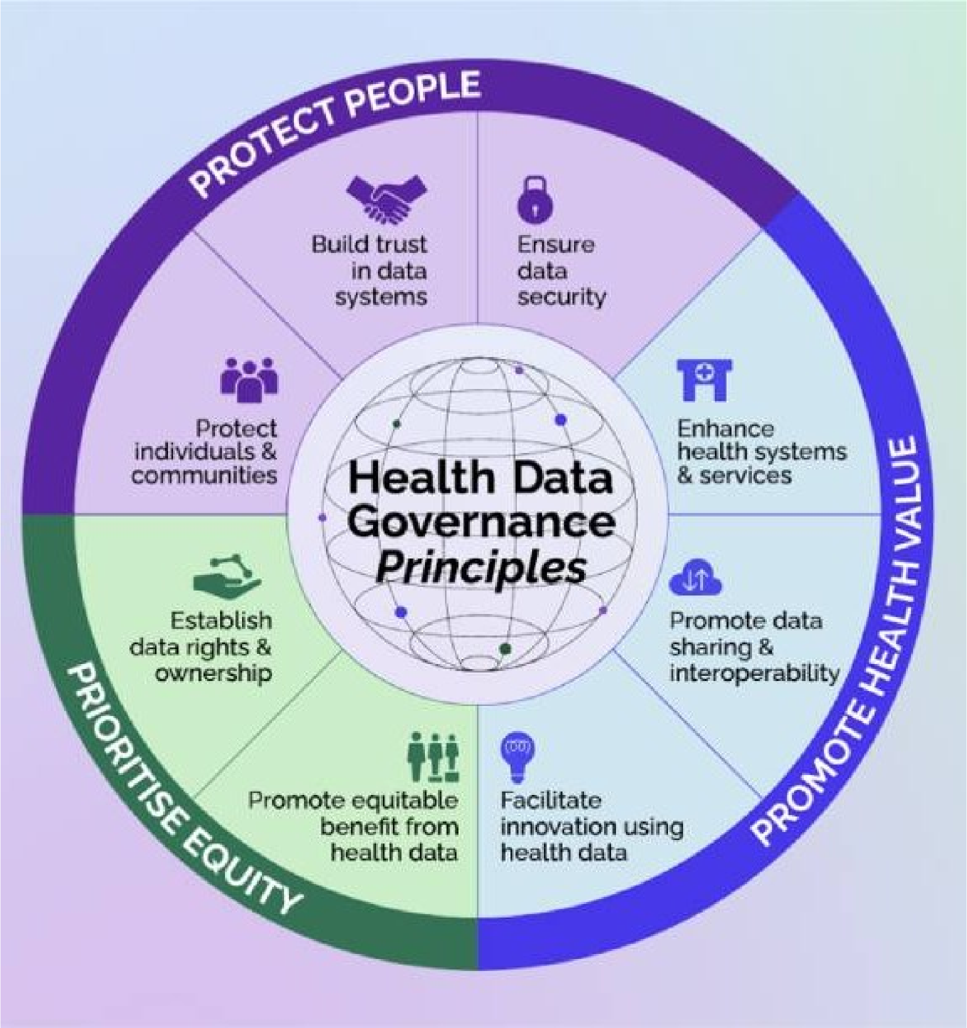
Embedded methods

- Perform feature selection as part of the model training process, with the algorithm automatically identifying the most important features.
- Examples: LASSO (shrinks less important variables to zero), decision trees (select important splits)

Filter methods

- Rank features using statistical criteria independently of the model and select the top-performing variables before training.
- Examples: Correlation tests, chi-square tests

Data Governance



- Protect People
  - Maintain confidentiality and privacy
  - Avoid stigma or harm from data misuse
  - Ensure sensitive conditions (e.g., HIV, TB) are handled carefully
  - Be transparent about how data is collected and used
  - Clearly communicate study purpose and benefits
  - Engage communities in research processes
  - Protect data from breaches and unauthorised access
  - Use encryption and secure servers
  - Restrict access to authorised researchers only

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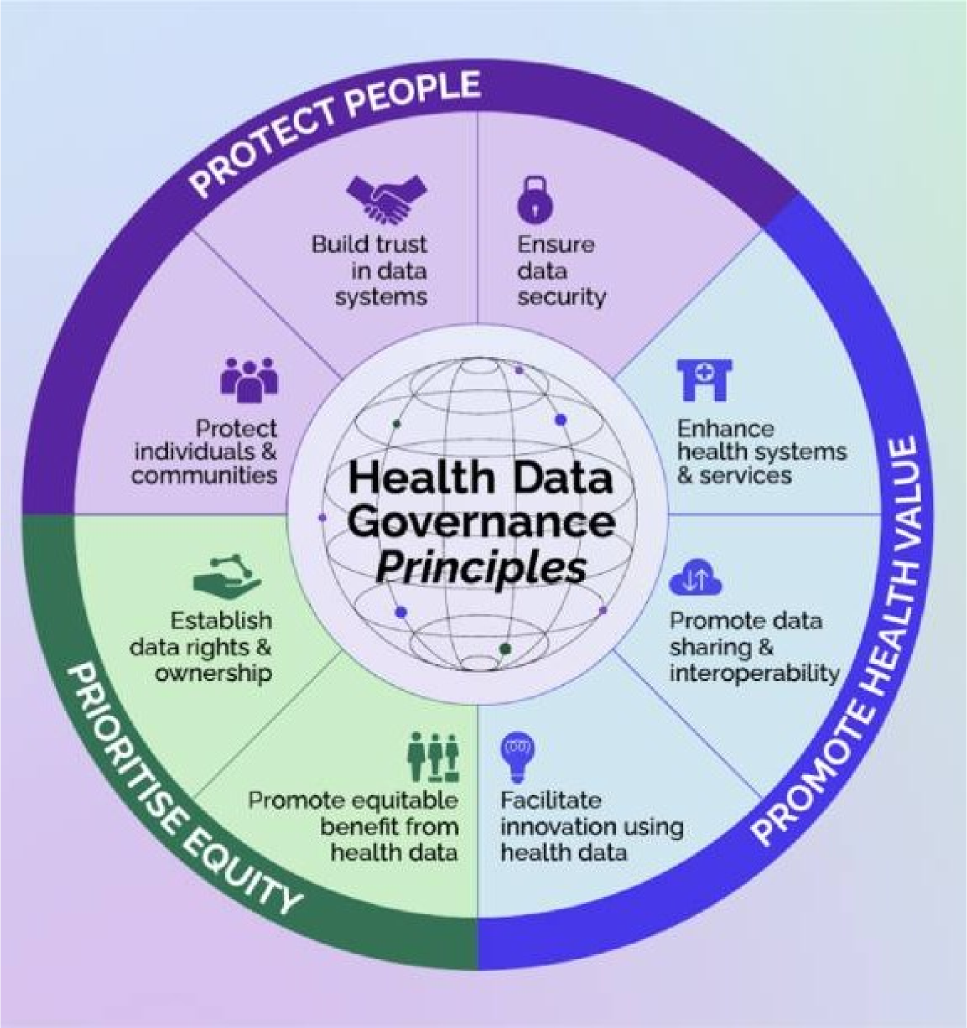

- Promote Health Value
  - Use data to improve patient care and outcomes
  - Inform clinical decision-making
  - Strengthen health planning
  - Enable systems to exchange data effectively
  - Use standard formats and coding systems
  - Break down data silos
  - Support research and machine learning applications
  - Enable discovery of new insights
  - Encourage responsible AI use

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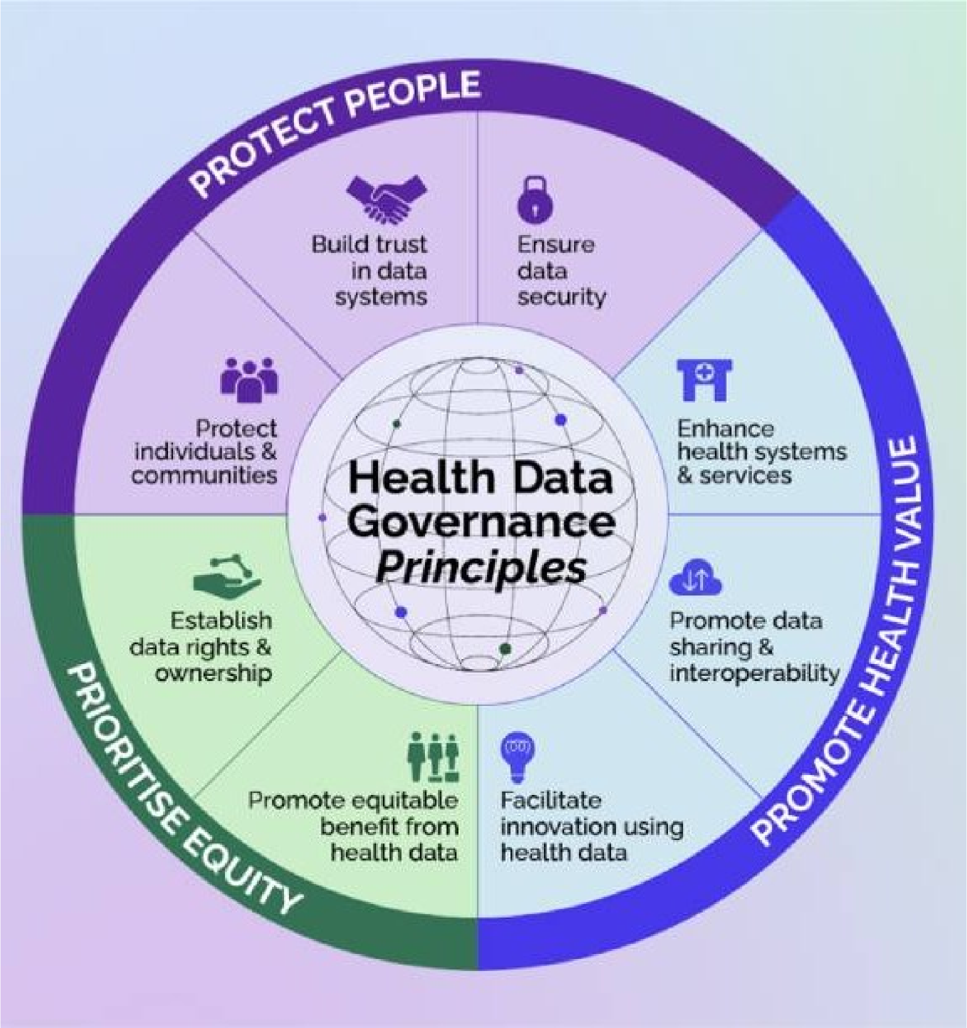

- Prioritise Equity
  - Ensure all populations benefit from insights
  - Avoid exclusion of vulnerable groups
  - Address health disparities
  - Define who owns and controls health data
  - Ensure ethical data access and use
  - Respect patient rights

ML Models

Labelled data → Supervised learning model → Predicted label

Unlabelled data → Unsupervised learning model → Clusters

Labelled and unlabelled data → Semi-supervised learning model → Predicted label

Machine Learning Algorithms

(Every data scientist must know)